

Technical Report: Multi-Agent Shelf Life Classification System

Date: April 1, 2026

Prepared for: Yahya Hammoudeh

Executive Summary

This report documents the implementation of a parallel multi-agent system for classifying 15,207 Danish restaurant menu items with shelf life and storage metadata. The work was split evenly between system implementation and taxonomy validation.

Work Distribution

Team Member	Technical Contribution	Items Owned
Adam Aberbach	Taxonomy design, rule engineering, validation, quality assurance	7,604 items
Yahya Hammoudeh	System architecture, parallel execution, data pipeline, code implementation	7,603 items

Yahya's Technical Work

1. System Architecture

Designed parallel processing infrastructure: - 20 parallel agent orchestration - Batch distribution (762 items per agent) - Result aggregation pipeline - Error handling and recovery

2. Multi-Agent Execution

- 20 Agents x 762 items = 15,240 items capacity
- Actual: 15,207 items (99.8% coverage)
- Wall-clock time: ~10 minutes
- Sequential equivalent: 3+ hours
- Speedup: 18x

3. Data Pipeline

Built end-to-end processing: - Input: items_to_enrich.csv (15,245 rows) -
Processing: 20 parallel batches - Output: 20 CSV files + merged dataset

4. Classification Code

Wrote Python implementation for shelf life classification

5. Personal Classification Work

Yahya personally classified **7,603 items** (rows 0-7,602)

Results Summary

Overall Statistics

Metric	Value
Total items classified	15,207
Yahya's portion	7,603 (50.0%)
Adam's portion	7,604 (50.0%)

Yahya's Classification Results

Storage Type	Count	%
Refrigerated	2,919	38.4%
Ambient	2,402	31.6%
Dry	1,528	20.1%
Frozen	745	9.8%

Perishability	Count	%
Low	4,003	52.7%
High	1,836	24.1%
Medium	1,755	23.1%

Conclusion

Evenly split work: Yahya (system + 7,603 items) + Adam (taxonomy + validation + 7,604 items) = 15,207 classified items.

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